UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	:	**COMPUTER ORGANIZATION AND ARCHITECTURE**
COURSE CODE	:	**CSC429**
EXAMINATION	:	**JUNE 2019**
TIME	:	**3 HOURS**

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions)
   PART B (1 Question)

2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack consists of :

   i) the Question Paper
   ii) an Answer Booklet – provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 7 printed pages

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PART A (80 MARKS)

QUESTION 1

Question 1 is based on the following Figure 1.

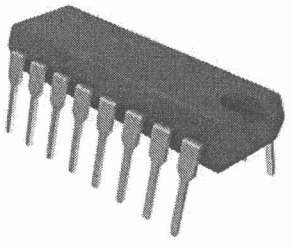


Figure 1: Component X

The development of component **X** led to a drastic increase in speed and efficiency of computers. Component **X** is a single, complete electronic semiconductor circuits contained on pieces of silicon, and sometimes called as chips.

a) What is the name of component X?

(1 mark)

b) In which computer generation was component X introduced?

(1 mark)

c) Who is the inventor of component X?

(1 mark)

d) State **ONE (1)** input device and **ONE (1)** output device that was introduced in the same generation as component X.

(2 marks)

QUESTION 2

The Von Neumann Architecture was first described in a seminal *First Draft of a Report on the EDVAC*. The document describes a design architecture of a general-purpose electronic computer that serves as the basis for almost all of modern computers.

a) List **TWO (2)** main components of a Von Neumann Architecture.

(2 marks)

b) Von Neumann computers are general purpose computers. Define the term *general-purpose computers*.

(2 marks)

c) Name **ONE (1)** of the earliest computers built based on the Von Neumann Architecture.

(1 mark)

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QUESTION 3

a) Convert the following values to the respective number bases. Show the complete work flow.

i) 10111.101_2 to decimal
ii) 507.875_{10} to hexadecimal

(5 marks)

b) Calculate $988_{16} - AB_{16}$ using the following techniques. Show the complete work flow.

i) Sign and magnitude
ii) 2's complement

(5 marks)

QUESTION 4

a) Convert -176.8125_{10} to floating point number in IEEE single precision format in hexadecimal value form.

(5 marks)

b) Convert $3F00C000_{16}$ in IEEE single precision floating point number to decimal value form.

(5 marks)

QUESTION 5

a) Registers are one of the main components of a CPU. Define the term registers, and list **TWO (2)** types of *registers.*

(4 marks)

b) Briefly explain the *Fetch* and *Decode* operations of a Machine Cycle.

(4 marks)

c) Buses are the most common type of system interconnection. Discuss **ONE (1)** difference between a *dedicated* bus and *multiplexed* bus.

(2 marks)

QUESTION 6

a) Assume that a portion of tags in the cache looks like Table 1. Identify the specific block address location for the following addresses:

i) EEE876
ii) ABE89B
iii) CDE891

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Table 1: Portion of Tags in Cache

Tag (binary)	Line number (binary)	Address with block			
		00	01	10	11
1100 1101	1110 1000 1001 00	A1	A2	A3	A4
1100 1100	1110 1000 1001 11	B1	B2	B3	B4
1010 1011	1110 1000 1001 10	C1	C2	C3	C4
1010 1010	1110 1000 0111 00	D1	D2	D3	D4
1110 1111	1110 1000 0111 10	E1	E2	E3	E4
1110 1110	1110 1000 0111 01	F1	F2	F3	F4

(3 marks)

b) What is the type of mapping function used in Table 1?

(1 mark)

c) Name and describe **ONE (1)** replacement algorithm that can be applied to both associative and set associative mapping function.

(3 marks)

d) State the effects towards memory's *Access Time, Cost* and *Capacity* as it goes down the memory hierarchy.

(3 marks)

QUESTION 7

Question 7 is based on the following scenarios.

- While cleaning his house, Ahmad found an old cassette tape of a popular 80's band. He played the tape and listened to several songs according to the listed tracks.
- Hafiz was tasked to research on the history of Malay culture by his professor. He went to the National Archive to search for information and manage to access one the resources which was stored in a floppy disk.
- Sara is planning to buy a new computer. She aims to get one with good processing speed. While researching for computer specifications, she found out that larger cache space has a direct effect towards processing performance.
- Hana loves playing role-playing games (RPG) and is planning to buy a newly released game that requires a high-speed RAM. But, she is devastated when she found out that her game console does not support the minimum RAM requirement of the game.

a) Based on the above scenarios, identify the access method for each storage devices/memory mentioned in (i), (ii), (iii) and (iv).

(4 marks)

b) Describe *SRAM* and *DRAM*.

(4 marks)

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c) Which level of cache is the fastest in a three-level cache architecture? Which type of RAM is used to implement the cache?

(2 marks)

QUESTION 8

a) Discuss **TWO (2)** functions of an input/output module.

(4 marks)

b) External input/output devices can be broadly categorized into three categories. List **TWO (2)** of the categories and provide an appropriate example of devices for each categories.

(4 marks)

c) Name **ONE (1)** input/output technique that requires active CPU intervention and **ONE (1)** technique that does not require CPU intervention.

(2 marks)

QUESTION 9

Based on the following information given in Figure 2, fill Table 2 with values that will be loaded into AC if different addressing modes are used for instruction LOAD 300.

(5 marks)

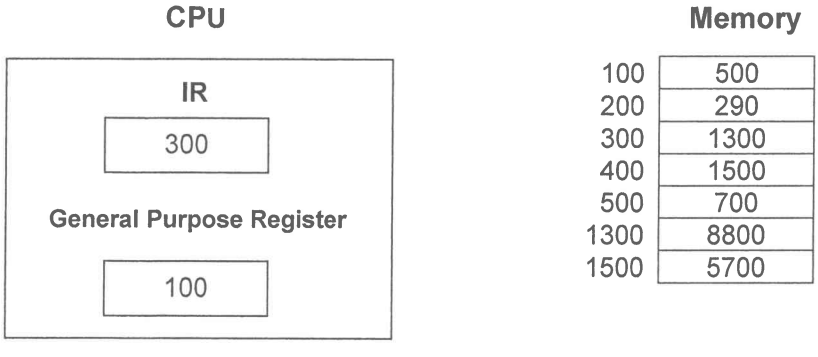


Figure 2: Different values in CPU and memory

Table 2: Values for Different Modes of Addressing

Mode of Addressing	Value
Immediate	a)
Direct	b)
Indirect	c)
Register	d)
Indirect Register	e)

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QUESTION 10

a) Describe **ONE (1)** major difference between a tightly coupled multiprocessor system and a loosely coupled multiprocessor system.

(2 marks)

b) State the **THREE (3)** types of cluster architecture.

(3 marks)

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PART B (20 MARKS)

QUESTION 1

Question 1 is based on Figure 3 and Table 3.

$$X = [(AB^2) + (A - BC)] / D$$

Figure 3: Equation X

Write the appropriate programs using assembly language notation to compute the equation represented in Figure 3 using **one-address**, **two-address**, and **three-address** instructions.

The instructions available for use are provided in Table 3. Note that register ***A, B, C*** and ***D*** cannot be modified and register ***x, y,*** and ***z*** can be used as temporary register. Final answer must be stored in register ***x***.

Table 3: Machine Instructions

One-address	Two-address	Three-address
LOAD m	MOV (m,n)	ADD (m,n,o)
STOR m	ADD (m,n)	SUB (m,n,o)
ADD m	SUB (m,n)	MUL (m,n,o)
SUB m	MUL (m,n)	DIV (m,n,o)
MUL m	DIV (m,n)	
DIV m		

(20 marks)

END OF QUESTION PAPER

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